

matrices approximates the attention matrix well. For low-entropy distributions (more peaked), sparse matrices are better-suited.

This implies that sparse and low-rank approximations could be complementary: if we can combine the strength of both, it is possible to come up with a better approximation across more general scenarios. Therefore, in the next observation, we try to combine sparse and low-rank approximations.

B.4 Observation 3: Sparse + Low-rank achieves better approximation error than sparse or low-rank alone

We find an approximation of the attention matrix of the form $S + L$, where S is sparse and L is low-rank. This problem has a rich history and is commonly solved with Robust PCA. As shown in 7, across the range of entropy, sparse + low-rank approximation can achieve lower error than either sparse or low-rank when choosing the correct mix ratio of sparse and low rank approximation ideally (with robust-PCA).

Motivated by the fact that sparse and low-rank approximations of attention matrices have complementary strengths (Observations 1 and 2), one might want to combine them (Observation 3) in hope of yielding a more robust approximation that works well across different kinds of attention matrices. The above introduces three main challenges that we have addressed in the main paper:

- how to find sparse + low-rank decomposition of an attention matrix that is compute efficient (the most studied algorithm, robust PCA, is orders of magnitude too slow to be done at each training iteration) and memory efficient (i.e., without materializing the full matrix) (Section 4),
- if we can find such a sparse + low-rank decomposition, how accurate is the approximation (Section 4.3),
- how expressive is the sparse + low-rank parameterization, i.e., are there natural classes of matrices where sparse + low-rank yields asymptotically better approximation than sparse or low-rank alone) (Section 3)?

C Scatterbrain Algorithm and Implementation Details

Let $Q, K \in \mathbb{R}^{n \times d}$ be the query and key matrices respectively, and $V \in \mathbb{R}^{n \times d}$ be the value matrix. Let the rows of Q be $q_1, \dots, q_n$, and the rows of K be $k_1, \dots, k_n$. The attention computes:

$$\text{softmax}(QK^\top)V,$$

with softmax applied row-wise, where for each vector $v \in \mathbb{R}^n$, $\text{softmax}(v) = \frac{1}{\sum_{j=1}^n e^{v_j}} [e^{v_1}, \dots, e^{v_n}]^\top$. Here we omit the usual scaling of $\frac{QK^\top}{\sqrt{d}}$ for simplicity since that could be folded into Q or K . Note that $\text{softmax}(QK^\top) = D^{-1} \exp(QK^\top)$, where the exponential function is applied element-wise and D is a diagonal matrix containing the softmax normalization constants ($D_{i,i} = \sum_{j=1}^n \exp(q_i^\top k_j)$). Then attention has the form $D^{-1} \exp(QK^\top)V$.

We describe the Scatterbrain approximation algorithm in Algorithm 1. This includes the normalization step.

Algorithm 1 Scatterbrain Approximation of Attention

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1: Input:  $Q, K, V \in \mathbb{R}^{n \times d}$ , hyper-parameters  $m, k, l$ 
2: procedure INIT( $m, k, l$ )
3:   Sample  $W \in \mathbb{R}^{m \times d}$  where  $W_i \sim \mathcal{N}(0, 1)$  i.i.d.
4:   Kernels  $\phi: \mathbb{R}^d \mapsto \mathbb{R}^m$ ,  $\phi(x) = \frac{\exp(Wx - \|x\|^2/2)}{\sqrt{m}}$ 
5:   Hash  $\forall l \in [L]$ ,  $\mathcal{H}_l = \{h_{l,k}\}_{k \in [K]}$ ,  $\mathcal{H} = \cup_{l \in [L]} \mathcal{H}_l$ 
6: end procedure
7: procedure LOWRANKAPPROX( $Q, K, V, \phi$ )
8:    $\tilde{Q} = \phi(Q)$ ,  $\tilde{K} = \phi(K)$  ▷ applied to each row
9:   return  $\tilde{Q}(\tilde{K}^\top V)$ ,  $\tilde{Q}(\tilde{K}^\top)1_n$ .
10: end procedure
11: procedure SPARSEAPPROX( $Q, K, V, \phi, \mathcal{H}$ )
12:    $\mathcal{S} = \{(i, j) | \mathcal{H}(Q_i) = \mathcal{H}(K_j)\}$ 
13:    $S \leftarrow$  sparse matrix whose support is  $\mathcal{S}$ 
14:   for  $(i, j) \in \mathcal{S}$  do
15:      $S_{ij} = \exp(q_i^\top k_j) - \phi(q_i)^\top \phi(k_j)$ .
16:   end for
17:   return  $SV$ ,  $S1_n$ .
18: end procedure
19: procedure SCATTERBRAINAPPROX( $Q, K, V$ )
20:    $\phi, h \leftarrow$  INIT( $m, k, l$ ).
21:    $O_{lr}, D_{lr} \leftarrow$  LOWRANKAPPROX( $Q, K, V, \phi$ ).
22:    $O_s, D_s \leftarrow$  SPARSEAPPROX( $Q, K, V, \phi, h$ ).
23:   return  $\text{diag}(D_{lr} + D_s)^{-1}(O_{lr} + O_s)$ .
24: end procedure

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Autoregressive / Causal / Unidirectional Attention To approximate autoregressive attention, we simply use the autoregressive variant of low-rank attention, and apply the autoregressive mask to the sparse attention. In particular, let $M \in \mathbb{R}^{n \times n}$ be the autoregressive mask, whose lower triangle is all ones and the rest of the entries are zero. The unnormalized attention matrix is $\exp((QK^\top) \odot M)$, and the unnormalized output is $\exp((QK^\top) \odot M)V$, where $\odot$ is elementwise multiplication.

The low-rank autoregressive variant computes $((\tilde{Q}\tilde{K}^\top) \odot M)V$, though with a custom GPU kernel / implementation so as not to materialize the $n \times n$ matrix. For the sparse component, we simply mask out locations S_{ij} where $i > j$. That is, we can perform $S \odot M$ efficiently. As a result, we can compute the Scatterbrain output $((\tilde{Q}\tilde{K}^\top) \odot M)V + (S \odot M)V$ efficiently.

D Proofs

D.1 Expressiveness of Sparse + Low-rank Matrices

To motivate the use of sparse + low-rank matrices, we describe a family of attention matrices where sparse + low-rank matrices need asymptotically fewer parameters to approximate the attention matrix, compared to sparse or low-rank matrices alone. For these cases, either sparse or low-rank alone requires a quadratic number of parameters ($O(n^2)$, where $n \times n$ is the dimension of the attention matrix) to get ϵ approximation error in Frobenius norm, while sparse + low-rank only requires $O(n)$ parameters.

We construct a matrix family that shows the separation between the approximation capability of sparse + low-rank vs. sparse or low-rank alone. More specifically, we will use diagonal + low-rank (a special case of sparse + low-rank).

Example 1. Let ϵ denote a parameter that satisfies $\epsilon \in (0, 1/2]$. Consider the following randomized construction of a matrix $Q \in \mathbb{R}^{n \times d}$ with $d \geq 6\epsilon^{-2} \log n$ and $d = \Theta(\epsilon^{-2} \log n)$, where each entry of Q is picked independently and uniformly at random from $\{\pm 1/\sqrt{d}\}$. Let $M = \sigma(QQ^\top)$ where σ is the elementwise exponential function (we first ignore the normalization term of softmax here).

It can be shown (e.g. by Hoeffding's inequality) that with high probability

$$(QQ^\top)_{i,j} = \begin{cases} 1, & \text{if } i = j; \\ \in [-\epsilon, \epsilon], & \text{otherwise.} \end{cases}$$

Since $M = \sigma(QQ^\top)$ where σ is the elementwise exponential function,

$$M_{i,j} = \begin{cases} e, & \text{if } i = j; \\ \in [1 - O(\epsilon), 1 + O(\epsilon)], & \text{otherwise.} \end{cases}$$

Intuitively, as the attention matrix M has large diagonal entries, low-rank matrices will not be able to approximate it well. However, the off-diagonals are also of reasonable size, thus making sparse approximation difficult. With sparse + low-rank, we can use the sparse part to represent the diagonal, and the low-rank part to represent the remaining elements, allowing it to approximate this matrix well. We formalize this separation in the theorem below.

Theorem 3. Let M be the attention matrix from Example 1. For any $\gamma \in [0, 1]$, with probability at least $1 - n^{-1}$, there exists a sparse + low-rank estimator with $O(\gamma^{-1} n^{3/2} \log n)$ parameters that achieve $\gamma\sqrt{n}$ Frobenius error. For any matrix $R \in \mathbb{R}^{n \times n}$ with rank such that $n - \text{rank} = \Omega(n)$ (e.g., R has $o(n^2)$ parameters), with probability at least $1 - n^{-1}$, we have $\|M - R\|_F \geq \Omega(\sqrt{n})$. Moreover, any matrix E_S that has row sparsity k (each row has less than k non-zeros) such that $n - k = \omega(1)$ (e.g., E_S has $o(n^2)$ parameters) will have error $\|M - E_S\|_F \geq \Omega(\sqrt{n})$ with probability at least $1 - n^{-1}$.

We see that for any $\gamma \in [0, 1]$, any low-rank or sparse estimator for M with (n^2) parameters has $\Omega(\gamma^{-1})$ times the error of the sparse + low-rank estimator with $O(\gamma^{-1} n^{1.5} \log n)$ parameters.

Proof of Theorem 3. For each $i \in [n]$, let q_i denote the i -th row of $Q \in \mathbb{R}^{n \times d}$. Define $J \in \mathbb{R}^{n \times n}$ to be the all 1s matrix. Define $T = M - J - QQ^\top$. Therefore,

$$T_{i,j} = \begin{cases} e - 2 & \text{if } i = j \\ e^{q_i^\top q_j} - 1 - q_i^\top q_j & \text{otherwise} \end{cases}.$$

By Hoeffding's inequality, for a pair $i \neq j$, we have that

$$\mathbb{P}(|q_i^\top q_j - \mathbb{E}[q_i^\top q_j]| \geq \epsilon) \leq 2 \exp \left(- \frac{2\epsilon^2}{\left(\frac{1}{\sqrt{d}} - \frac{-1}{\sqrt{d}}\right)^2} \right) = 2 \exp(-d\epsilon^2/2).$$

Note that $\mathbb{E}[q_i^\top q_j] = 0$.

By a union bound over all pairs $i \neq j$ (there are $n(n-1)/2$ such pairs), with probability at least $1 - n^2 \exp(-d\epsilon^2/2)$, we have that

$$q_i^\top q_j \in [-\epsilon, \epsilon] \quad \text{for all } i \neq j.$$

Since we assume that $d \geq 6\epsilon^{-2} \log n$, we have that

$$n^2 \exp(-d\epsilon^2/2) \leq n^2 \exp(-3 \log n) = n^{-1}.$$

Hence $q_i^\top q_j \in [-\epsilon, \epsilon]$ for all $i \neq j$ with probability at least $1 - n^{-1}$. For the rest of the proof, we only consider this case (where $q_i^\top q_j \in [-\epsilon, \epsilon]$ for all $i \neq j$).

Since $1 + x \leq e^x \leq 1 + x + x^2$ for $|x| < 1$, we can bound the off diagonal elements $|T_{i,j}| \leq \epsilon^2$. In particular, for all $i \neq j$,

$$|T_{i,j}| = \left| e^{q_i^\top q_j} - 1 - q_i^\top q_j \right| \leq (q_i^\top q_j)^2 \leq \epsilon^2. \quad (4)$$

Sparse + low-rank estimator: We use the following sparse + low-rank estimator:

$$E_{\text{SL}} = \underbrace{(e-2) \cdot I}_{\text{sparse}} + \underbrace{J + QQ^\top}_{\text{low-rank}},$$

where $(e-2)I$ has row sparsity 1 and $\text{rank}(J + QQ^\top) \leq d+1 = O(\epsilon^{-2} \log n)$.

Notice that the E_{SL} estimate matches M exactly on the diagonal, and on the off-diagonal it differs from M by $T_{i,j}$. Thus, the Frobenius error of the sparse + low-rank estimator is

$$\|M - E_{\text{SL}}\|_F \leq \epsilon^2 \sqrt{n(n-1)} \leq \epsilon^2 n.$$

Set $\epsilon = \frac{\sqrt{\gamma}}{n^{1/4}}$ for $0 \leq \gamma \leq 1$, Then

- (i) The sparse + low-rank parameter count is $n + n \cdot \text{rank} \leq n \cdot O(\epsilon^{-2} \log n) \leq O(\gamma^{-1} n^{1.5} \log n)$.
- (ii) The Frobenius error is $\leq \gamma \sqrt{n}$.

Low-rank estimator: We want to argue that low-rank approximation would require more parameters. If we approximate the matrix $(e-2)I$ by a matrix R with rank r , then the difference matrix will have at least $n-d$ singular values of magnitude $e-2 \geq 1/2$. As a result, by the Eckart–Young–Mirsky theorem,

$$\|(e-2) \cdot I - R\|_F \geq \frac{1}{2} \sqrt{n-r}.$$

Define $T' = T - (e-2) \cdot I$, then T' is all 0 on the diagonal and has absolute value $\leq \epsilon^2$ on off-diagonal entries. Thus $\|T'\|_F \leq \epsilon^2 n = \gamma \sqrt{n}$.

We want to show that if R' is a rank r' matrix, then $\|M - R'\|_F \geq \frac{1}{2} \sqrt{n-r'-d-1} - \|T'\|_F$. We argue by contradiction. Suppose that there exists some matrix R' with rank r' such that

$$\|M - R'\|_F \leq \frac{1}{2} \sqrt{n-r'-d-1} - \|T'\|_F.$$

Define $R = R' - J - QQ^\top$, so $M - R' = (e-2) \cdot I - R + T'$. We see that:

$$\begin{aligned} \|(e-2) \cdot I - R\|_F &= \|M - R' - T'\|_F \\ &\leq \|M - R'\|_F + \|T'\|_F \\ &\leq \frac{1}{2} \sqrt{n-r'-d-1} \\ &\leq \frac{1}{2} \sqrt{n - \text{rank}(R)}. \end{aligned}$$

This contradicts the result above, which states that $\|(e-2) \cdot I - R\|_F \geq \frac{1}{2} \sqrt{n - \text{rank}(R)}$.

Therefore any low-rank estimator with rank r such that $n-r = \Omega(n)$, which has $\Omega(n^2)$ parameters, will have error at least $\Omega(\sqrt{n-r-d-1}) - \|T'\|_F = \Omega(\sqrt{n})$, which is $\Omega(\gamma^{-1})$ times the error of the sparse + low-rank estimator above.